Drifts Under Time-Varying Fields

Time Varying B Field

We know that a time varying magnetic field induces an electric field.

$$\nabla \times \vec{E} = -\frac{\partial B}{\partial t} \quad \text{or} \quad \oint \vec{E}\cdot d\vec{l} = -\int_S \dot{B}\cdot ds = -\frac{d\phi}{dt}$$

Thus a particle in a time-varying magnetic field will gain energy. The work done on the particle will be:

$$\delta W = -\oint |q|\vec{E}\cdot d\vec{l} = +|q|\int_S \dot{\vec{B}}\cdot d\vec{s} = +|q|\frac{d\phi}{dt} = |q|\dot{B}\pi r_L^2$$

We can then use energy-conservation to relate the change in kinetic energy to the work done.

$$\delta\left(\frac{1}{2}mv_\perp^2\right) = |q|\dot{B}\pi r_L^2 = \frac{2\pi\dot{B}m}{|q|B}\,\frac{\frac{1}{2}mv_\perp^2}{B} = \frac{2\pi\dot{B}}{|\Omega|}\mu$$

We will look at how this drift looks in a bit, for now let us look at how magnetic moment and flux can be conserved.

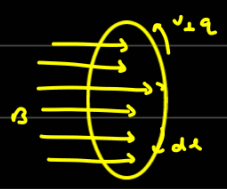
$$\frac{d}{dt}\left(\frac{1}{2}mv_\perp^2\right) = \frac{|\Omega|}{2\pi}\delta\left(\frac{1}{2}mv_\perp^2\right) = \mu\frac{dB}{dt} \quad \text{Also} \quad \frac{d}{dt}\left(\frac{1}{2}mv_\perp^2\right) = \frac{d}{dt}(\mu B)$$

This all implies that: $\frac{d\mu}{dt} = 0 \longrightarrow$ This implies conservation of magnetic moment.

The flux is given by: $\phi = \frac{2\pi m}{q^2}\mu$

Conservation of magnetic moment thus implies conservation of magnetic flux.

Now, let's analyze the drift.



Since flux is conserved, the net flux has to be 0. Thus if the magnetic field and thus the magnetic flux change with time, the particle will undergo a motion of a current carrying loop to oppose the change in flux.

If the particle was in a larmor gyration, it will simply or slower since we have an induced EMF.

Time Varying E-Field

We know that a uniform electric field causes the "E×B" drift:

$$v_{E\times B} = \frac{\vec{E}\times\vec{B}}{B^2}$$

Now, if the electric field varies with time, the $\vec{E}\times\vec{B}$ drift will also vary with time. This creates an acceleration.

$$\frac{d\dot{v}_{E\times B}}{dt} = \frac{d}{dt}\left(\frac{\vec{E}\times\vec{B}}{B^2}\right)$$

Now, using Newton's second law and the frame of the guiding centre, we can calculate the force felt by the particle.

$$F = -m\frac{d}{dt}\left(\frac{\vec{E}\times\vec{B}}{B^2}\right)$$ → This force is retarding in nature. Since we are in the frame of the guiding centre, this is like a pseudo-force.

This force will produce another drift called the polarization drift.

$$\vec{v}_p = \frac{1}{q}\frac{\vec{F}\times\vec{B}}{B^2} = \frac{m}{qB^2}\frac{d}{dt}\left(\frac{\vec{E}\times\vec{B}}{B^2}\right)\times\vec{B} = -\frac{m}{qB}\frac{d}{dt}\left((\vec{E}\cdot\vec{B})\vec{B} - B^2\vec{E}\right) = \frac{m}{qB^2}\dot{E}_\perp$$

The polarization drift will be along the ⊥electric field whenever it is changing w.r.t time.

The total drift will then be:

$$v_D = v_{E\times B} + v_p = \frac{\vec{E}\times\vec{B}}{B^2} + \frac{m}{qB^2}\dot{E}_\perp$$

Polarization Drift is the cause of the start-up effect. When an electric field is "turned on", the gyrocentre of a particle moves along the electric field for a bit until there is no change in the field.

The total shift is given by:

$$\delta\vec{r} = \int\vec{v}_p\,dt = \frac{m}{qB^2}\int\dot{E}_\perp\,dt = \frac{m}{qB^2}[\Delta E_\perp]$$

The derivation of the total field drift done above is only an approximation. This is because we are not accounting for the change of $v_{E\times B}$ with time. Let's try a more direct derivation now.

Consider an oscillatory field $\vec{E} = \vec{E}e^{-i\omega t}$ ($\perp r_0\vec{B}$)

$$m\frac{d\vec{v}}{dt} = q(\vec{E} + \vec{v}\times\vec{B}) = q(\vec{E}e^{-i\omega t} + \vec{v}\times\vec{B})$$

We can try for a solution of the form: $\vec{v} = \vec{v}_D e^{-i\omega t} + \vec{v}_L$ → Larmor velocity

$$m\frac{d}{dt}(\vec{v}) = m\frac{d}{dt}(\vec{v}_D e^{-i\omega t} + \vec{v}_L) = m(-i\omega v_D) \quad ; \; q(\vec{E} + \vec{v}_D\times\vec{B})e^{-i\omega t} + m\dot{v}_L$$

$$-mi\omega\vec{v}_D = q(\vec{E} + \vec{v}_D\times\vec{B})e^{-i\omega t} \longrightarrow m^2\omega^2\vec{v}_D = [mi\omega q\vec{E} + mi\omega q(\vec{v}_D\times\vec{B})]e^{-i\omega t}$$

Taking cross-product with B will give us:

$$-mi\omega(\vec{v}_D\times\vec{B}) = q(\vec{E}\times\vec{B} + (\vec{v}_D\times\vec{B})\times\vec{B})e^{-i\omega t} \longrightarrow -mi\omega q(\vec{v}_D\times\vec{B}) = q^2(\vec{E}\times\vec{B} - B^2\vec{v}_D)e^{-i\omega t}$$

$$= q(\vec{E}\times\vec{B} + (\vec{B}^2\cdot\vec{v}_D)\vec{B} - B^2\vec{v}_D)e^{-i\omega t}$$

Since we are equating the non-time varying part of the equation, we can ignore the exponentials to get:

$$m^2\omega^2 \vec{v}_D + q^2(\vec{E}\times\vec{B} - B^2\vec{v}_D) = mi\omega q\vec{E}$$

This gives us:

$$v_D\left[1-\frac{\omega^2}{\Omega^2}\right] = \frac{-i\omega q}{\Omega B|q|}\vec{E} + \frac{\vec{E}\times\vec{B}}{B^2}$$ → This is ALMOST the same formula as before since $-i\omega \longleftrightarrow \frac{\partial}{\partial t}$

The difference here is the $[1-\omega^2\Omega^2]$ term. This accounts for the change in v_D with time.

The old formula will be a good approximation iff $\omega \ll \Omega$

Drifts in Non-Uniform $\vec{E}$

We have the differential equation: $m\frac{d\vec{v}}{dt} = q(\vec{E}(r) + \vec{v}\times\vec{B})$

We can take the usual solution: $\vec{v}_D = \vec{v}_D + \vec{v}_g$ and average out over a gyro-orbit:

$$\left\langle m\frac{dv_D}{dt}\right\rangle = 0 = q[\langle\vec{E}(r)\rangle + \vec{v}_D\times\vec{B}] \longrightarrow \vec{v}_D = \frac{\langle\vec{E}(r)\rangle\times\vec{B}}{B^2}$$ → This is essentially a variant of the $E\times B$ drift.

The problem essentially boils down to finding the average field over a gyro orbit. To do so, we can expand $\vec{E}$ as a Taylor series about the GC:

$$\vec{E}(r) = \vec{E}_0 + (\vec{r}.\nabla)\vec{E} + \left(\frac{x^2}{2!}\frac{\partial^2}{\partial x^2} + \frac{y^2}{2!}\frac{\partial^2}{\partial y^2}\right)\vec{E} + \text{cross terms} + \ldots$$

↳ cross-terms average out to be 0.

Then we get:

$$\langle E(r)\rangle = \vec{E} + (\langle r_L\rangle.\nabla)\vec{E} + \frac{\langle r_L^2\rangle}{2!}\nabla^2\vec{E}$$

↳ 0's out

$$\therefore \langle E(r)\rangle \simeq \vec{E} + \frac{r_L^2}{4}\nabla^2 E$$

Thus we get $\vec{E}\times\vec{B}$ drift with the first Larmor radius correction:

$$\vec{v}_{E\times B} = \left(1 + \frac{r_L^2}{r}\nabla^2\right)\frac{\vec{E}\times\vec{B}}{B^2}$$ → Notice that the Grad B drift is a finite Larmor effect.